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Edited by

Howard S. Cohl and Mourad E. H. Ismail



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Preface

On July 11–15, 2016, we organized a summer school, *Orthogonal Polynomials and Special Functions*, Summer School 6 (OPSF-S6) which was hosted at the University of Maryland, College Park, Maryland. This summer school was co-organized with Kasso Okoudjou, Professor and Associate Chair, Department of Mathematics, and Norbert Wiener Center for Harmonic Analysis and Applications. OPSF-S6 was a National Science Foundation (NSF) supported summer school on orthogonal polynomials and special functions which received partial support from the Institute for Mathematics and its Applications (IMA), Minneapolis, Minnesota. Twenty-two undergraduates, graduate students, and young researchers attended the summer school from the USA, China, Europe, Morocco and Tunisia, hoping to learn a new state of the art in these subject areas.

Since 1970, the subjects of *special functions* and special families of *orthogonal polynomials*, have gone through major developments. The Wilson and Askey–Wilson polynomials paved the way for a better understanding of the theory of hypergeometric and basic hypergeometric series and shed new light on the pioneering work of Rogers and Ramanujan. This was combined with advances in the applications of q -series in number theory through the theory of partitions and allied subjects. When quantum groups arrived, the functions which appeared in their representation theory turned out to be the same q -functions which were recently developed at that time. This motivated researchers to revisit the old Bochner problem of studying polynomial solutions to second order differential, difference, or q -difference equations, which are of Sturm–Liouville type and have polynomial coefficients. This led to a generalization where having a solution of degree n for all $n = 0, 1, 2, \dots$ is now weakened to having a solution of degree n for all $n = N, N + 1, \dots$, for some finite N . The polynomial solutions are required to be orthogonal. This gave birth to the subject of

exceptional orthogonal polynomials. What is amazing is that these orthogonal polynomials continued to be complete in the L_2 space weighted by their orthogonality measure. At the same time, harmonic analysis of the newly developed functions attracted the attention of many mathematicians with different backgrounds. Alongside the analytic theory of the newly discovered functions and polynomials, a powerful combinatorial theory of orthogonal polynomials evolved. This also included a combinatorial theory of continued fractions. In the late 1990s, the elliptic gamma function was introduced and the theory of elliptic special functions followed. This area overlaps with mathematical physics because there are physical models whose solutions use elliptic special functions. The theory of multivariate orthogonal polynomials and special functions, as well as the harmonic analysis of root systems was also developed. The Selberg integral from 1944 became a prime example of integrals over root systems.

In this summer school, we felt that we needed to cover topics of current interest, as well as newly evolving topics which are expected to evolve and blossom into very active areas of research. This is hard to do in a classical subject like special functions. We also tried to cover areas within the broad scope of this classical and well established subject which have not been covered in recent summer schools. The fruits of our labor are in the lecture notes written by the lecturers of the summer school contained in this volume. We believe these notes are detailed enough to learn or teach graduate level classes from, or cover in advanced seminars. They are full of insights into the subjects covered, and are written by experts whose research is at the leading edge of the subject.

The lecturers were Antonio Durán (Departamento de Análisis Matemático, Universidad de Sevilla, Sevilla, Spain), Mourad Ismail (Department of Mathematics, University of Central Florida, Orlando, Florida, USA), Erik Koelink (Department of Mathematics, Radboud Universiteit Nijmegen, Nijmegen, The Netherlands), Hjalmar Rosengren (Chalmers University of Technology and University of Gothenburg, Gothenburg, Sweden), and Jiang Zeng (Institut Camille Jordan, Université Claude Bernard Lyon 1, Villeurbanne, Lyon, France).

The first set of these lecture notes is by Antonio Durán who has made significant contributions in the subject of matrix orthogonal polynomials and exceptional orthogonal polynomials. Both subjects are relatively new, but their roots go back to the 1940s and have evolved very rapidly since the 1990s. These notes give a very nice introduction to the subject of classical orthogonal polynomials, and covers some very recent results on two expansions of the Askey tableau: Krall and exceptional polynomials; both

in their continuous and discrete versions. Critical to Durán's discussion is the theory of $\mathcal{D}$ -operators, how they can be used to construct Krall polynomials, and the use of duality to construct exceptional discrete polynomials. Durán clearly explains the relations between all of these families via limiting processes and using Christoffel transformations of the measure (Darboux transformations). Examples are treated in detail including the orthogonal polynomial families of Krall–Laguerre, Krall–Meixner, Krall–Charlier, exceptional Hermite, exceptional Laguerre, exceptional Meixner, and exceptional Charlier polynomials. These notes discuss the above developments and describes as well the current state of the art.

Mourad Ismail's lectures represent a nice tutorial which gives a brief review of many useful and recent results in the theory of q -series. He uses the Askey–Wilson operators, polynomial expansions, and analytic continuation to derive many of the fundamental identities and transformations of q -series. The Askey–Wilson calculus plays a major role in his approach, and even though this approach was also used in his 2005 book, our understanding of these analytic and more conceptual techniques has clearly advanced in the last few years. Some of these techniques were already used for the development of the theory of hypergeometric functions on root systems. This approach is more conceptual than the classical approach presented in most books on the subject. We expect this tutorial to be an advantageous text for learners of the subject.

The lectures by Erik Koelink cover the spectral theory of self-adjoint operators related to, or arising from, problems involving special functions and orthogonal polynomials. Of paramount importance is the spectral decomposition (the spectral theorem) of compact bounded and unbounded self-adjoint operators on infinite-dimensional Hilbert spaces. In particular, the emphasis in these notes is on the spectral theory of Jacobi operators, and especially on Jacobi operators corresponding to matrix-valued orthogonal polynomials. The theory of such Jacobi operators is built up, and several explicit examples are presented. In addition, he surveys the rigorous mathematical treatment of the J -matrix method and the more general concept of tridiagonalization. The J -matrix method was originally introduced by physicists and has been applied very successfully to many quantum mechanical systems.

Hjalmar Rosengren's lectures provides an excellent introduction to elliptic hypergeometric functions, a relatively recent class of special functions. These lecture notes fill a hole in the demand for a clear introduction to the field. He dedicates the first part of his lecture to a review of important topics connected to the subject, such as additive and multiplicative

elliptic functions, theta functions, factorization, Weierstrass's three-term identity, even elliptic functions, partial fraction decomposition, modularity and elliptic curves. Throughout his lectures, through examples and a host of carefully constructed exercises, he connects his discussion with important and illustrative facts in complex function theory and to generalized and basic hypergeometric functions. He closes the lectures with a highly relevant historical application from statistical mechanics which is how elliptic hypergeometric series first appeared in the literature. This application concerns fused Boltzmann weights for Baxter's solvable elliptic solid-on-solid lattice model.

Note that in the abstract of his lectures, Rosengren says "By contrast, there is no hint about elliptic hypergeometric series in the literature until 1988." This statement is definitely true, but Mourad Ismail would like to shed some light on this issue. In 1982, he visited the Chudnovsky brothers and Gregory Chudnovsky asked him to explore orthogonal polynomials whose three-term recurrence coefficients are elliptic functions. Gregory also said that an elliptic generalization of the continuous q -ultraspherical polynomials should exist. Mourad was unable to solve this problem, but Gregory Chudnovsky anticipated the *elliptic revolution* in hypergeometric functions.

It is our opinion that a mathematical theory is best understood when formulated at the correct level of generality. This is indeed the case with the one variable theory of special functions. It is conceptually simpler at the most general level, the elliptic level.

The lecture notes by Jiang Zeng report on some recent topics developed in the cross-cutting field of combinatorics, special functions and orthogonal polynomials. This subject is relatively recent, although it is rooted in earlier works from the 1940s by Touchard, Kaplansky, Riordan, and Erdős. Zeng presents side-by-side, the two dual combinatorial approaches to orthogonal polynomials, namely that of Flajolet and Viennot, and that of Foata.

The lectures begin by reviewing essential results in the theory of general orthogonal polynomials, namely Favard's theorem, three-term recurrence relations and their connection to continued fraction expansions through generating functions for the moments. Since, orthogonal polynomials can be considered as enumerators of finite structures with respect to some statistics, Zeng illustrates how to use combinatorial models to explore orthogonal polynomial identities. As examples, Zeng develops combinatorial models for orthogonal Sheffer polynomials (Hermite, Laguerre, Charlier, Meixner, and Meixner–Pollaczek) through exponential generating func-

tions, and for Al-Salam–Chihara polynomials (and their rescaled special sub-families including continuous q -Hermite, q -Charlier, and q -Laguerre polynomials) to combinatorially compute their moments and linearization coefficients. Zeng gives some details about Ismail, Stanton and Viennot’s groundbreaking combinatorial evaluation of the Askey–Wilson integral and inquires whether there may exist combinatorial proofs of the Nassrallah–Rahman integral and Askey–Wilson moments. There are even recent applications of this approach to the asymmetric simple exclusion process (ASEP), where the moments of the Askey–Wilson polynomials play a central role.

Organizing such a summer school was a pleasure, although we encountered many challenges along the way. Kasso Okoudjou led the local organizational efforts for OPSF-S6. It was a pleasure working with Kasso throughout the entire process, both before and after the summer school. He skillfully maneuvered all aspects of the summer school, including the successful procurement of NSF and IMA funding, without which OPSF-S6 would not have been nearly as successful as it was. Kasso smoothly interacted with graduate students, early career researchers, and lecturers, for travel planning, booking, and ultimately for reimbursement. It was both an honor and a pleasure working with Kasso whose influence and organization with the Norbert Wiener Center for Harmonic Analysis and Applications is greatly appreciated. The lecture rooms we used, that Kasso obtained, were both useful and comfortable; snacks and coffee were always available; and the local staff was extremely pleasant and efficient. We would also like to acknowledge the support of Dan Lozier and Stephen Casey for their help and advice in planning the summer school.

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Exceptional Orthogonal Polynomials via Krall Discrete Polynomials

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Abstract: We consider two important extensions of the classical and classical discrete orthogonal polynomials: namely, Krall and exceptional polynomials. We also explore the relationship between both extensions and how they can be used to expand the Askey tableau.

Introduction

In these lectures, we will consider two important extensions of the classical and classical discrete orthogonal polynomials: namely, Krall and exceptional polynomials. On the one hand, Krall or bispectral polynomials are orthogonal polynomials that are also common eigenfunctions of higher-order differential or difference operators. On the other hand, exceptional polynomials have recently appeared in connection with quantum mechanical models associated to certain rational perturbations of classical potentials. We also explore the relationship between both extensions and how they can be used to expand the Askey tableau.

Section 1.1. *Background on classical and classical discrete polynomials.*

The explicit solution of certain mathematical models of physical interest often depends on the use of special functions. In many cases, these special functions turn out to be certain families of orthogonal polynomials which, in addition, are also eigenfunctions of second-order operators of some specific kind. We can consider these families as the workhorse of all classical mathematical physics, ranging from potential theory, electromagnetism, etc. through the successes of quantum mechanics in the 1920s in the hands of Schrödinger. These families are called “classical orthogonal polynomials”. E.J. Routh proved in 1884 (see [77]) that the only families of orthogonal polynomials (with respect to a positive weight) that can

be simultaneous eigenfunctions of a second-order differential operator are those going with the names of Hermite, Laguerre and Jacobi. This result is also a consequence of the Bochner classification theorem of 1929 [7]. The similar question for second-order difference operators gave rise to the classical discrete families of orthogonal polynomials (Charlier, Meixner, Krawtchouk, Hahn), classified by Lancaster in 1941 [61]. Finally, second-order q -difference operators gave rise to the q -classical families of orthogonal polynomials (Askey–Wilson, q -Racah, etc.), although the q -families will not be considered in these lectures.

Sections 1.2, 1.3 and 1.4. *The Askey tableau. Constructing Krall and Krall discrete orthogonal polynomials using $\mathcal{D}$ -operators.*

Since all these families of orthogonal polynomials can be represented by hypergeometric functions, they are also known as hypergeometric orthogonal polynomials, and they are organized as a hierarchy in the so-called Askey tableau.

As an extension of the classical families, more than 75 years ago the first families of orthogonal polynomials which are also eigenfunctions of higher-order differential operators were discovered by H.L. Krall, who classified orthogonal polynomials which are also eigenfunctions of differential operators of order 4. Because of that, orthogonal polynomials which are eigenfunctions of differential or difference operators of higher order are usually called Krall polynomials and Krall discrete polynomials, respectively. Following the terminology of Duistermaat and Grünbaum (see [11]), they are also called bispectral, and this is because of the following reason. In the continuous parameter, they are eigenfunctions of the above-mentioned operators, and in the discrete parameter, they are eigenfunctions of a second-order difference operator: the three-term recurrence relation (which is equivalent to orthogonality with respect to a measure supported in the real line).

Since the 1980s, a lot of effort has been devoted to studying Krall polynomials, with contributions by L.L. Littlejohn, A.M. Krall, J. Koekoek and R. Koekoek. A. Grünbaum and L. Haine (and collaborators), K. H. Kwon (and collaborators), A. Zhedanov, P. Iliev, and many others. The orthogonality of all these families is with respect to particular cases of Laguerre and Jacobi weights together with one or several Dirac deltas (and its derivatives) at the end points of their interval of orthogonality.

Surprisingly enough, until very recently no example was known in the case of difference operators, despite the problem being explicitly posed

by Richard Askey in 1990, more than twenty five years ago. The results known for the continuous case do not provide enough help: indeed, adding Dirac deltas to the classical discrete weights does not seem to work. The first examples of Krall discrete polynomials appeared three years ago in a paper of mine, where I proposed some conjectures on how to construct Krall discrete polynomials by multiplying the classical discrete weights by the “annihilator polynomial” of certain finite sets of numbers. These conjectures have been already proved by using a new concept: $\mathcal{D}$ -operators. $\mathcal{D}$ -operators provide a unified approach to construct Krall, Krall discrete or q -Krall orthogonal polynomials.

This approach has also led to the discovery of new and deep symmetries for determinants whose entries are classical and classical discrete orthogonal polynomials, and has led to an unexpected connection of these symmetries with Selberg type formulas and constant term identities.

Sections 1.5 and 1.6. *Exceptional orthogonal polynomials. The dual connection with Krall polynomials at the discrete level. Expanding the Askey tableau.*

Exceptional orthogonal polynomials are complete orthogonal polynomial systems with respect to a positive measure which in addition are eigenfunctions of a second-order differential operator. They extend the classical families of Hermite, Laguerre and Jacobi.

The most apparent difference between classical orthogonal polynomials and exceptional orthogonal polynomials is that the exceptional families have a finite number of gaps in their degrees. That is, not all degrees are present in the sequence of polynomials (as it happens with the classical families). Besides that, they form a complete orthonormal set in the underlying Hilbert space defined by the orthogonalizing positive measure.

This means in particular that they are not covered by the hypotheses of Bochner’s classification theorem. Each family of exceptional polynomials is associated to a quantum-mechanical potential whose spectrum and eigenfunctions can be calculated using the exceptional family. These potentials turn out to be, in each case, a rational perturbation of the classical potentials associated to the classical polynomials. Exceptional orthogonal polynomials have been applied to shape-invariant potentials, supersymmetric transformations, discrete quantum mechanics, mass-dependent potentials, and quasi-exact solvability. Exceptional polynomials appeared some nine years ago, but there has been a remarkable activity around them mainly by theoretical physicists (with contributions by D. Gómez-Ullate,

N. Kamran and R. Milson, Y. Grandati, C. Quesne, S. Odake and R. Sasaki, and many others).

In the same way, exceptional discrete orthogonal polynomials are an extension of discrete classical families such as Charlier, Meixner, Krawtchouk and Hahn. They are complete orthogonal polynomial systems with respect to a positive measure (but having gaps in their degrees) which in addition are eigenfunctions of a second-order difference operator.

Taking into account these definitions, it is scarcely surprising that no connection has been found between Krall and exceptional polynomials. However, if one considers Krall discrete polynomials, something very exciting happens: duality (roughly speaking, swapping the variable with the index) interchanges Krall discrete and exceptional discrete polynomials. This unexpected connection of Krall discrete and exceptional polynomials allows a natural and important extension of the Askey tableau. Also, this worthy fact can be used to solve some of the most interesting questions concerning exceptional polynomials; for instance, to find necessary and sufficient conditions such that the associated second-order differential operators do not have any singularity in their domain. This important issue is very much related to the existence of an orthogonalizing measure for the corresponding family of exceptional polynomials.

1.1 Background on classical and classical discrete polynomials

1.1.1 Weights on the real line

These lectures deal with some relevant examples of orthogonal polynomials. Orthogonality here is with respect to the inner product defined by a weight in the linear space of polynomials with real coefficients (which we will denote by $\mathbb{R}[x]$).

Definition 1.1 A weight μ is a positive Borel measure with support in the real line satisfying

- (1) μ has finite moments of every order: i.e., the integrals $\int x^n d\mu(x)$ are finite for $n \in \mathbb{N} := \{0, 1, 2, \dots\}$;
- (2) μ has infinitely many points in its support (x is in the support of μ if $\mu(x - \varepsilon, x + \varepsilon) > 0$ for all $\varepsilon > 0$).

Condition (2) above is equivalent to saying that

$$\int p^2(x) d\mu(x) > 0, \quad \text{for } p \in \mathbb{R}[x], p \neq 0. \quad (1.1)$$

We associate to the weight μ the inner product defined in $\mathbb{R}[x]$ by

$$\langle p, q \rangle_\mu = \int p(x)q(x)d\mu(x). \quad (1.2)$$

By applying the Gram–Schmidt orthogonalization process to the sequence $1, x, x^2, x^3, \dots$ (condition (1.1) is required), one can generate a sequence $(p_n)_n$, $n = 0, 1, 2, 3, \dots$, of polynomials with $\deg p_n = n$ that satisfy the orthogonality condition

$$\int p_n(x)p_m(x)d\mu(x) = c_n\delta_{n,m}, \quad c_n > 0.$$

We then say that the polynomial sequence $(p_n)_n$ is orthogonal with respect to the weight μ . When $c_n = 1$, we say that the polynomial p_n is orthonormal. It is not difficult to see that the orthogonal polynomial p_n is unique up to multiplicative constants.

Except for multiplicative constants, the n th orthogonal polynomial p_n is characterized because $\int p_n(x)q(x)d\mu(x) = 0$ for $q \in \mathbb{R}[x]$ with $\deg q \leq n - 1$. This is trivially equivalent to saying that $\int p_n(x)x^k d\mu(x) = 0$ for $0 \leq k \leq n - 1$.

Condition (2) above can be weakened; if we assume that the positive measure μ has N points in its support (say, x_i , $1 \leq i \leq N$), with $N < \infty$, we can only guarantee the existence of N orthogonal polynomials p_n , $n = 0, 1, \dots, N - 1$, with respect to μ . Indeed, up to multiplicative constants, the polynomial $p(x) = \prod_{i=1}^N (x - x_i)$ is the only one with degree N orthogonal to p_n , $n = 0, 1, \dots, N - 1$. But the norm of p is zero.

1.1.2 The three-term recurrence relation

Orthogonality with respect to a weight can be characterized in terms of an algebraic equation. Indeed, since xp_n is a polynomial of degree $n + 1$, we can expand it in terms of the polynomials $p_0, p_1, \dots, p_{n+1}$:

$$xp_n(x) = a_{n+1}p_{n+1}(x) + b_n p_n(x) + c_n p_{n-1}(x) + d_n p_{n-2}(x) + \dots + e_n p_0(x). \quad (1.3)$$

Using the orthogonality of the polynomial p_n to the polynomials of lower degree, we have

$$\begin{aligned} d_n \langle p_{n-2}(x), p_{n-2}(x) \rangle_\mu &= \langle xp_n(x), p_{n-2}(x) \rangle_\mu = \int_{\mathbb{R}} xp_n(x)p_{n-2}(x)d\mu \\ &= \langle p_n(x), xp_{n-2}(x) \rangle_\mu = 0. \end{aligned}$$

That is $d_n = 0$. Proceeding in a similar way, we deduce that in the expansion (1.3), only the polynomials p_{n+1} , p_n and p_{n-1} appear:

$$xp_n(x) = a_{n+1}p_{n+1}(x) + b_n p_n(x) + c_n p_{n-1}(x). \quad (1.4)$$

It is not difficult to see that the positivity of μ implies that $a_n c_n > 0$. For the orthonormal polynomials we have the symmetry condition $c_n = a_n$.

The converse of (1.4) is also true and it is the spectral theorem for orthogonal polynomials (it is also known as Favard's theorem [32], although the result seems to be known already to Stieltjes, Chebyshev, and others, and is contained in the book by Stone [81] which appeared a couple of years before Favard's paper).

Theorem 1.2 *If the sequence of polynomials $(p_n)_n$ with $\deg p_n = n$ satisfies the three-term recurrence relation (1.4) with $a_n c_n > 0$, then they are orthogonal with respect to a weight.*

For a proof see [3], [10] or [81].

1.1.3 The classical orthogonal polynomial families

The most important examples of orthogonal polynomials are the so-called classical families. There are three such families (see, for instance, [10, 31, 50, 53, 67, 82], and for a good historical account see [4, 10]).

- (1) The Jacobi polynomials $(P_n^{(\alpha, \beta)})_n$, $\alpha, \beta > -1$. They are a double parametric family of orthogonal polynomials with respect to the weight $d\mu = (1-x)^\alpha (1+x)^\beta dx$ on the interval $(-1, 1)$. There are some relevant particular values of the parameters which deserve special interest. The simplest case is when $\alpha = \beta = 0$ which results in Legendre polynomials. Legendre introduced them at the end of the 18th century and they are the first example of orthogonal polynomials in history (more about that later). The cases $\alpha = \beta = -1/2$ and $\alpha = \beta = 1/2$ are called Chebyshev polynomials of the first and second kind, respectively, and they were introduced and studied by this Russian mathematician in the second half of the 19th century. When $\alpha = \beta$ we have Gegenbauer or ultraspherical polynomials.
- (2) The Laguerre polynomials $(L_n^\alpha)_n$, $\alpha > -1$. They are a single parametric family of orthogonal polynomials with respect to the weight $d\mu(x) = x^\alpha e^{-x} dx$ on the half line $x > 0$.

- (3) The Hermite polynomials $(H_n)_n$. They are a single family of orthogonal polynomials with respect to the weight $d\mu(x) = e^{-x^2}dx$ on the real line.

I would like to note that, as usual in mathematics, many of the orthogonal families which appear in these lectures are misnamed, in the sense that someone else introduced them earlier than the person after whom the family is named. The classical families are related to the three most important continuous distribution of probabilities: Beta, Gamma and Normal, respectively. The classical families enjoy a set of important characterization properties that we will discuss in later sections.

There are many other examples of orthogonal polynomials. For instance, the so-called Heine polynomials, a single parametric family of orthogonal polynomials with respect to the weight $d\mu(x) = [x(1-x)(a-x)]^{-1/2}dx$ on the interval $(0, 1)$ where $a > 1$ [10]. Of course there are sequences of orthogonal polynomials with no name: as far as I know the orthogonal polynomials with respect to the weight $d\mu(x) = |\cos(x^2 + 1)|dx$ on the interval $(0, 2)$ have not yet been baptized. However, any careful reader will be concerned by the previous examples because actually they are not examples of orthogonal polynomials: they are just examples of weights!

Orthogonal polynomials can be explicitly computed only for a few weights. *Explicitly* means here a closed-form expression for each of the polynomials p_n in the family. For an arbitrary weight (as the one with no name above) one can hardly compute an approximation of the first few orthogonal polynomials. The classical families are among those happy cases in which we can compute *everything* explicitly (most of the identities we will consider next can be found in many books, see, for instance, [10, 31, 50, 53, 67, 82]).

To start with, let us consider the Legendre polynomials $(P_n)_n$, i.e., the Jacobi case with $\alpha = \beta = 0$. They are orthogonal with respect to the Lebesgue measure on the interval $[-1, 1]$. Here is a hint for finding explicitly the Legendre polynomials: try with the differentiation formula

$$P_n(x) = ((1-x)^n(1+x)^n)^{(n)}. \quad (1.5)$$

Trivially we get $\deg P_n = n$. We then have to prove that

$$\int_{-1}^1 P_n(x)x^k dx = 0, \quad 0 \leq k \leq n-1.$$

This can be easily proved using the following integration by parts:

$$\begin{aligned}
 & \int_{-1}^1 ((1-x)^n (1+x)^n)^{(n)} x^k dx \\
 &= ((1-x)^n (1+x)^n)^{(n-1)} x^k \Big|_{-1}^1 - k \int_{-1}^1 ((1-x)^n (1+x)^n)^{(n-1)} x^{k-1} dx \\
 &= -k \int_{-1}^1 ((1-x)^n (1+x)^n)^{(n-1)} x^{k-1} dx \\
 &= \dots = (-1)^k k! \int_{-1}^1 ((1-x)^n (1+x)^n)^{(n-k)} dx \\
 &= (-1)^k k! ((1-x)^n (1+x)^n)^{(n-1-k)} \Big|_{x=-1}^{x=1} = 0.
 \end{aligned}$$

Formula (1.5) is called the Rodrigues' formula for the Legendre polynomials, honoring the French mathematician Olinde Rodrigues who discovered it at the beginning of the 19th century.

Each classical family of orthogonal polynomials has a corresponding Rodrigues' formula of the form

$$p_n(x) = (a_2^n(x) \mu(x))^{(n)} / \mu(x), \quad (1.6)$$

where a_2 is a polynomial of degree at most 2 and μ is the corresponding weight function for the family. More explicitly

(1) Jacobi polynomials

$$P_n^{(\alpha, \beta)}(x) = \frac{(-1)^n}{2^n n!} ((1-x^2)^n (1-x)^\alpha (1+x)^\beta)^{(n)} (1-x)^{-\alpha} (1+x)^{-\beta}.$$

(2) Laguerre polynomials

$$L_n^\alpha(x) = \frac{1}{n!} (x^n x^\alpha e^{-x})^{(n)} e^x x^{-\alpha}.$$

(3) Hermite polynomials

$$H_n(x) = (-1)^n (e^{-x^2})^{(n)} e^{x^2}.$$

One of the characteristic properties of the classical families is precisely this kind of Rodrigues' formula. These Rodrigues' formulas allow an explicit computation for each family. For instance, one get for the Legendre and Laguerre polynomials, respectively

$$\begin{aligned}
 P_n(x) &= \frac{1}{2^n} \sum_{k=0}^{[n/2]} (-1)^k \binom{n}{k} \binom{2n-2k}{n} x^{n-2k}, \\
 L_n^\alpha(x) &= \sum_{k=0}^n \binom{n+\alpha}{n-k} \frac{(-x)^k}{k!}.
 \end{aligned} \quad (1.7)$$

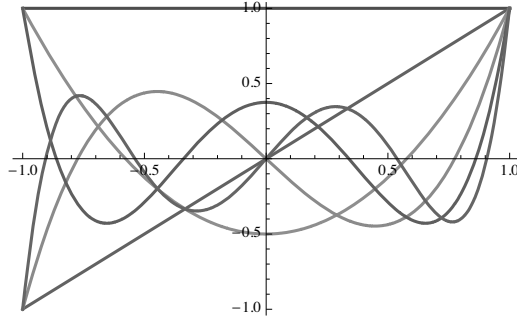


Figure 1.1 The first few Legendre polynomials

Exercise 1.3 Prove the explicit expression for the Laguerre polynomials.

Using the explicit formulas for the Legendre polynomials one can draw the first few of them, as shown in Figure 1.1.

One can then check that each one of these first few Legendre polynomials has real and simple zeros which live in the interval of orthogonality $(-1, 1)$. Actually this is a general property of the zeros of orthogonal polynomials.

Theorem 1.4 *The zeros of an orthogonal polynomial p_n with respect to a weight μ are real, simple and live in the convex hull of the support of μ .*

Exercise 1.5 Prove this theorem.

Hint: let $x_1, \dots, x_m$ be the real zeros of p_n with odd multiplicity. Consider then the polynomial $q(x) = (x - x_1) \cdots (x - x_m)$. Show that $p_n(x)q(x)$ has constant sign in $\mathbb{R}$. Use that fact and the orthogonality of p_n to conclude that $m \geq n$.

Rodrigues' formulas can also be used to explicitly compute the three-term recurrence relation (1.4). That is the case for the Hermite polynomials:

$$\begin{aligned} H_{n+1}(x) &= (-1)^{n+1} (e^{-x^2})^{(n+1)} e^{x^2} = (-1)^{n+1} (-2xe^{-x^2})^{(n)} e^{x^2} \\ &= (-1)^{n+1} \left(-2x(e^{-x^2})^{(n)} - 2n(e^{-x^2})^{(n-1)} \right) e^{x^2} \\ &= 2xH_n(x) - 2nH_{n-1}(x). \end{aligned}$$

The three-term recurrence relation for Laguerre and Jacobi polynomials

are, respectively

$$\begin{aligned}
 L_{n+1}^\alpha(x) &= \frac{1}{n+1} \left((2n+1+\alpha-x)L_n^\alpha(x) - (n+\alpha)L_{n-1}^\alpha(x) \right), \\
 2(n+1)(n+\alpha+\beta+1)(2n+\alpha+\beta)P_{n+1}^{(\alpha,\beta)}(x) \\
 &= -2(n+\alpha)(n+\beta)(2n+\alpha+\beta+2)P_{n-1}^{(\alpha,\beta)}(x) \\
 &\quad + (2n+\alpha+\beta+1)[(\alpha^2-\beta^2) \\
 &\quad + (2n+\alpha+\beta)(2n+\alpha+\beta+2)x]P_n^{(\alpha,\beta)}(x).
 \end{aligned}$$

1.1.4 Second-order differential operator

Legendre polynomials appeared in the context of planetary motion. Indeed, Legendre studied the following potential function related to planetary motion

$$V(t, y, z) = \iiint \frac{\rho(u, v, w)}{r} du dv dw,$$

where $r = \sqrt{(t-u)^2 + (y-v)^2 + (z-w)^2}$, and $\rho(u, v, w)$ stands for the density at the point (u, v, w) (Cartesian coordinates). This integral is easier after performing the change of variable

$$r(\rho, x) = (1 - 2\rho x + \rho^2)^{1/2}, \quad x = \cos \gamma.$$

Legendre then expanded $1/r(\rho, x)$ in power series of ρ

$$\frac{1}{(1 - 2\rho x + \rho^2)^{1/2}} = \sum_{n=0}^{\infty} \rho^n P_n(x), \quad (1.8)$$

and he realized that the functions $P_n(x)$ are actually polynomials in x of degree n : they are the polynomials we now call Legendre polynomials.

The function in the left-hand side of identity (1.8) is called the generating function for the Legendre polynomials. The generating function was an invention of the Swiss mathematician Leonard Euler: his genius idea was to pack a sequence $(a_n)_n$ in an analytic function f ,

$$f(z) = \sum_{n=0}^{\infty} a_n z^n,$$

so that one can find interesting properties of the sequence $(a_n)_n$ from the function f , as long as one can explicitly find the function f . As the identity (1.8) shows, that is the case of the Legendre polynomials. A very interesting property one can deduce for the Legendre polynomials from its generating function is that they satisfy a second-order differential equation.

Indeed, by a direct computation one can see that the function $f(\rho, x)$ in (1.8) satisfies the partial differential equation

$$(1-x^2)\frac{\partial^2}{\partial x^2}f(\rho, x) - 2x\frac{\partial}{\partial x}f(\rho, x) = -\rho\frac{\partial^2}{\partial \rho^2}(\rho f(\rho, x)).$$

Substituting the function $f(\rho, x)$ by its power expansion (1.8), one can straightforwardly find the following second-order differential equation for the Legendre polynomials

$$(1-x^2)P_n''(x) - 2xP_n'(x) = -n(n+1)P_n(x). \quad (1.9)$$

Since in this differential equation the coefficients of the second and first derivatives do not depend on n , the differential equation is actually the eigenfunction equation for the second-order differential operator

$$D = (1-x^2)\frac{\partial^2}{\partial x^2} - 2x\frac{\partial}{\partial x}.$$

The differential equation (1.9) can also be proved using the orthogonality of the Legendre polynomials. Indeed, write $q_n(x) = (1-x^2)p_n''(x) - 2xp_n'(x)$. Obviously $\deg q_n = n$. For $0 \leq k \leq n-1$, an integration by parts gives

$$\begin{aligned} & \int_{-1}^1 \left((1-x^2)P_n''(x) - 2xP_n'(x) \right) x^k dx \\ &= (1-x^2)P_n'(x)x^k \Big|_{-1}^1 - \int_{-1}^1 P_n'(x) \left(-2xx^k + kx^{k-1}(1-x^2) \right) dx \\ & \quad - \int_{-1}^1 P_n'(x) 2xx^k dx \\ &= -k \int_{-1}^1 P_n'(x) \left(x^{k-1}(1-x^2) \right) dx \\ &= -k(1-x^2)P_n(x)x^{k-1} \Big|_{-1}^1 \\ & \quad + k \int_{-1}^1 P_n(x) \left((k-1)x^{k-2}(1-x^2) - 2xx^{k-1} \right) dx \\ &= 0. \end{aligned}$$

That is, $(1-x^2)P_n''(x) - 2xP_n'(x)$ is orthogonal to the polynomials of degree at most $n-1$. Since orthogonal polynomials are unique up to multiplicative constants we get $(1-x^2)P_n''(x) - 2xP_n'(x) = \lambda_n P_n(x)$. Comparing leading coefficients in both sides of this equality, we recover the differential equation (1.9).

Each classical family of orthogonal polynomials is also the eigenfunctions of a second-order differential operator:

(1) Jacobi polynomials

$$\begin{aligned} (1-x^2) \left(P_n^{(\alpha, \beta)} \right)''(x) + (\beta - \alpha - (\alpha + \beta + 2)x) \left(P_n^{(\alpha, \beta)} \right)'(x) \\ = \lambda_n P_n^{(\alpha, \beta)}(x), \\ \lambda_n = -n(n + \alpha + \beta + 1). \end{aligned}$$

(2) Laguerre polynomials

$$x(L_n^\alpha)''(x) + ((\alpha + 1) - x)(L_n^\alpha)'(x) = -nL_n^\alpha(x). \quad (1.10)$$

(i) Hermite polynomials

$$H_n''(x) - 2xH_n'(x) = -2nH_n(x). \quad (1.11)$$

Being eigenfunctions of a second-order differential operator is another characteristic property of the classical families. Probably this is the most relevant of these properties if we take into account the important physical applications of these second-order differential equations (more about this later).

1.1.5 Characterizations of the classical families of orthogonal polynomials

The last characteristic property of the classical families considered in these lectures is the so-called Pearson's equation. The Rodrigues' formula (1.6) for $n = 1$ gives the following first-order differential equation for the classical weights

$$(a_2(x)\mu(x))' = a_1(x)\mu(x),$$

where a_i is a polynomial of degree at most i . This is the so-called Pearson equation for the classical weights, where the coefficients a_2 and a_1 are precisely the differential coefficients of the second-order differential operator for the corresponding classical family.

We summarize some of the characteristic properties of the classical families in the following theorem.

Theorem 1.6 *Let $(p_n)_n$ be a sequence of orthogonal polynomials with respect to a weight $d\mu$. Let a_2 and a_1 be polynomials of degree at most 2 and 1 respectively. Then the following conditions are equivalent:*

$$(1) \quad a_2(x)p_n''(x) + a_1(x)p_n'(x) = \lambda_n p_n(x), \quad n \geq 0.$$

- (2) The weight is absolutely continuous with respect to the Lebesgue measure $d\mu(x) = \mu(x)dx$ with μ an smooth function satisfying $(a_2\mu)' = a_1\mu$.
- (3) Up to a linear change of variable, $(p_n)_n$ are the classical families: Jacobi, Laguerre or Hermite.
- (4) The polynomials can be defined by means of the Rodrigues' formula $p_n(x) = (a_2^n \mu)^{(n)} / \mu$.

The implication (1) $\Rightarrow$ (3) in this theorem was proved by E.J. Routh in 1884 (see [77]), but it is known as the Bochner classification theorem (see Bochner's paper [7] of 1929). Actually Bochner classified polynomial solutions to Sturm–Liouville problems (which, beside the classical orthogonal polynomials, also include the solution $(x^n)_n$). For a complete treatment of this theorem we refer to [53] (see also [67]).

Exercise 1.7 Under the additional assumptions

- (1) $d\mu(x) = \mu(x)dx$, with μ a twice differentiable positive function supported on an interval (a, b) of the real line (not necessarily bounded),
- (2) $x^n a_2 \mu, x^n (a_2 \mu)'$ and $x^n a_1 \mu$ have vanishing limits at a and b ,

prove the implications (2) $\Rightarrow$ (1), (2) $\Rightarrow$ (4) and (2) $\Leftrightarrow$ (3).

Hint: The two first implications are similar to what we did for Legendre polynomials. The last one is a straightforward integration once one has reduced to the cases $a_2(x) = 1, x, 1 - x^2, x^2$, after a linear change of variable.

(The classical families satisfy other important characteristic properties that we do not consider in these lectures; for instance the sequence of derivatives $(p'_{n+1})_n$ is again orthogonal with respect to a positive measure (with respect to $a_2 \mu$)).

1.1.6 The classical families and the basic quantum models

The classical families of orthogonal polynomials appear when the Schrödinger equation is integrated in basic models for non-relativistic quantum mechanics (see [67]). Let $\psi(t, x) = \psi_t(x)$ be a wave function for a quantum system; then the Schrödinger equation is

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = H(\psi),$$

where H is the Hamiltonian of the system and $\hbar \equiv$ Planck constant. When the Hamiltonian H is independent of time, this equation can be reduced to

$$H(\phi)(x) = E\phi(x).$$

This equation is usually called the time-independent Schrödinger equation and coincides with the eigenvalue equation for the energy of the system, because the Hamiltonian H is the operator associated to the energy (so that the corresponding eigenvalue E gives the energy of the system).

Consider now the one-dimensional harmonic oscillator, i.e., the system consisting of a particle attracted to the origin with force proportional to the displaced distance: $F = -kx$ where k is a constant of the system (for instance, the system formed by a vibrating diatomic molecule). The potential is then $V(x) = -\int F(t)dt = \frac{1}{2}kx^2$ and the Hamiltonian is given by

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2}kx^2,$$

which is obviously independent of time, and where m is the particle mass.

The Schrödinger equation is then

$$-\frac{\hbar^2}{2m} \psi''(x) + \frac{1}{2}kx^2 \psi(x) = E \psi(x). \quad (1.12)$$

This equation has a square integrable solution only when

$$E_n = (2n+1) \frac{\hbar}{2} \sqrt{\frac{k}{m}},$$

which provides the spectrum of energies for the system.

The second-order differential equation (1.12) can then be reduced to the corresponding one for the Hermite polynomials (1.11), so that the wave functions for the different states of the system can be constructed by means of the Hermite polynomials

$$\psi_n(x) = \left(\frac{\sqrt{b}}{n!2^n} \right)^{\frac{1}{2}} H_n(\sqrt{b}x) e^{-\frac{b}{2}x^2}, \quad b = \frac{\sqrt{km}}{\hbar}.$$

Consider now the model for the hydrogen atom according to the Schrödinger equation, [79]. This is a three-dimensional model describing the interaction between the proton (at the nucleus) and the electron given by the Coulombian potential

$$V(x,y,z) = -\frac{Ze^2}{4\pi\epsilon_0} \frac{1}{\sqrt{x^2+y^2+z^2}},$$

with $Z \equiv$ atomic number and $e \equiv$ electron charge.

The equation was proposed and solved by Erwin Schrödinger. It gives the values of the energy spectrum found experimentally. Because of that, he won in 1933 the Nobel Physics prize. The Schrödinger equation for the hydrogen atom can be solved by separation of variables. One then gets a

second-order differential equation for the radial part and another second-order differential equation for the angular part. The radial equation can be reduced to the second-order differential equation for the Laguerre polynomials. In the angular equation the so-called spherical harmonics appear. They are given by

$$Y_n^\lambda(\theta, \rho) = N e^{i\lambda\rho} C_n^\lambda(\cos \theta),$$

where C_n^λ , $\lambda > -1/2$, $\lambda \neq 0$, stands for the n th ultraspherical polynomial (Jacobi family with $\alpha = \beta = \lambda - 1/2$).

For a trio of quantum numbers $n, l, m \in \mathbb{N}$, the wave function corresponding to this quantum state can then be constructed by using Laguerre and ultraspherical (Gegenbauer) polynomials; it has the form (spherical coordinates):

$$\psi_{n,l,m}(r, \theta, \rho) = A_{n,l,m} e^{-r/2} r^l L_{n+l}^{2l+1}(r) Y_l^m(\theta, \rho).$$

Most of the images for the hydrogen atom are drawn using this formula. For instance in the *Encyclopaedia Britannica* one can find the images, reproduced in Figure 1.2 for the electron density functions ((x, z) -plane) for the quantum numbers $n = 8$, $m = 0$ and $l = 0, 2, 6$ and 7 .

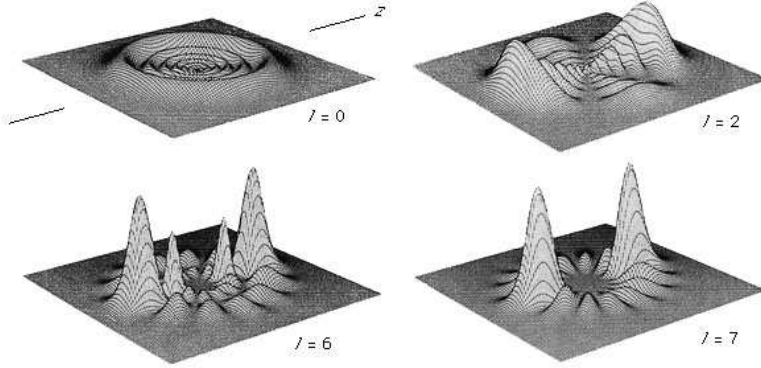


Figure 1.2

1.1.7 The classical discrete families

The classical discrete families follow the classical families in the ranking of the most important examples of orthogonal polynomials. Roughly speaking they are obtained by substituting derivatives with differences in the operational calculus of the classical families (most of the identities we

will consider next can be found in [10, 50, 53, 67]). In order to be more precise, we need some notation.

For $a \in \mathbb{R}$, we denote by $\mathfrak{s}_a$ the shift operator

$$\mathfrak{s}_a f(x) = f(x + a).$$

As usual Δ and ∇ denote the basic delta and nabla first-order difference operators

$$\Delta f = f(x+1) - f(x) = (\mathfrak{s}_1 - \text{Id})f, \quad \nabla f = f(x) - f(x-1) = (\text{Id} - \mathfrak{s}_{-1})f.$$

They are also known in the literature of finite differences as forward and backward difference operators.

A classical discrete family is formed by orthogonal polynomials that in addition are also eigenfunctions of a second-order difference operator with polynomial coefficients. We have several equivalent possibilities for writing such second-order difference operators:

$$\left. \begin{aligned} D &= a_2 \Delta^2 + a_1 \Delta + a_0, \\ D &= a_2 \Delta \nabla + a_1 \Delta + a_0, \\ D &= a_1 \mathfrak{s}_1 + a_0 \mathfrak{s}_0 + a_{-1} \mathfrak{s}_{-1}. \end{aligned} \right\} \quad (1.13)$$

We prefer to work with the last one. Since we want the operator D (1.13) to have a sequence of orthogonal polynomials as eigenfunctions, D has to satisfy the necessary condition $\deg(D(p)) \leq \deg(p)$ for $p \in \mathbb{R}[x]$. It is easy to see that this is equivalent to saying that the coefficients a_1 , a_0 and a_{-1} are polynomials of degree at most 2 satisfying $\deg(a_1 + a_0 + a_{-1}) = 0$ and $\deg(a_1 - a_{-1}) = 1$. Hence we implicitly assume along these lectures those assumptions for the coefficients of the second-order difference operator (1.13).

Lancaster classified in 1941 (see [61]) all families of orthogonal polynomials which, in addition, are eigenfunctions of a second-order difference operator of the form (1.13). He found four families of orthogonal polynomials which had already appeared in the literature. They are associated to the names of Charlier, Meixner, Krawtchouk and Hahn. They are known as the classical discrete families and are related with some of the most important discrete probability distributions: Poisson, Pascal, binomial and hypergeometric, respectively. More precisely:

- (1) Charlier polynomials $(c_n^a)_n$, $a > 0$, orthogonal with respect to (Poisson distribution)

$$\sum_{n=0}^{\infty} \frac{a^n}{n!} \delta_n. \quad (1.14)$$

- (2) Meixner polynomials $(m_n^{a,c})_n$, $0 < a < 1$, $c > 0$, orthogonal with respect to (Pascal distribution)

$$\sum_{n=0}^{\infty} \frac{a^n \Gamma(n+c)}{n!} \delta_n.$$

- (3) Krawtchouk polynomials $(k_n^{a,N})_{0 \leq n \leq N}$, $a > 0$, $N \in \mathbb{N}$, orthogonal with respect to (binomial distribution)

$$\sum_{n=0}^N a^n \binom{N}{n} \delta_n.$$

- (4) Hahn polynomials $(h_n^{a,b,N})_{0 \leq n \leq N}$, $a > 0$, $N \in \mathbb{N}$, orthogonal with respect to (hypergeometric distribution)

$$\sum_{n=0}^N \binom{a+n}{n} \binom{b+N-n}{N-n} \delta_n.$$

Notice that both Krawtchouk and Hahn polynomials are finite families.

All the classical discrete weights are discrete measures supported either in the non-negative integers (Charlier and Meixner) or the first $N+1$ non-negative integers (Krawtchouk and Hahn). The classical discrete families have relevant applications in many areas of mathematics and other sciences. For instance, one of the most important ball and urn models due to P. and T. Ehrenfest (1907) can be modeled by means of Krawtchouk polynomials (see [51], also [42]).

Let us recall that for a real number u the Dirac delta δ_u is the Borel measure defined by

$$\delta_u(A) = \begin{cases} 1, & u \in A, \\ 0, & u \notin A. \end{cases}$$

Hence, an integral with respect to δ_u just picks up the value of the corresponding function at u : $\int f d\delta_u = f(u)$. Hence, integrals with respect to the classical discrete weights reduce to sums. For instance if we write μ for the Charlier weight (1.14), we have

$$\int f d\mu = \sum_{n=0}^{\infty} \frac{a^n}{n!} f(n).$$

The classical discrete families enjoy a number of important characteristic properties, some of which are summarized in the following Theorem.

Theorem 1.8 *Let $(p_n)_n$ be a sequence of orthogonal polynomials with*

respect to a weight $d\mu = \sum_{n=0}^N \mu(n)\delta_n$, where N is infinite or a positive integer. Let a_1 , a_0 and a_{-1} be polynomials of degree at most 2 satisfying that $\deg(a_1 + a_0 + a_{-1}) = 0$, $\deg(a_1 - a_{-1}) = 1$. Then the following conditions are equivalent:

- (1) $a_1 p_n(x+1) + a_0 p_n(x) + a_{-1} p_n(x-1) = \lambda_n p_n(x)$.
- (2) $a_{-1}(x+1)\mu(x+1) = a_1(x)\mu(x)$, $x \in \{0, \dots, N-1\}$, with initial condition $a_{-1}(0) = 0$ and $a_1(N) = 0$ (if N is finite).
- (3) Up to a linear change of variable $(p_n)_n$ are some of the classical discrete families: Charlier, Meixner, Krawtchouk or Hahn.
- (4) $p_n(x) = \nabla^n(\prod_{k=1}^n a_{-1}(x+k)\mu(x+n))/\mu(x)$.

Implication (1) $\Rightarrow$ (2) is the Lancaster classification theorem already mentioned. For a complete treatment of this theorem we refer to [53] (see also [68]).

Exercise 1.9 Prove the implication (2) $\Rightarrow$ (1).

Hint: Show that the polynomial

$$q_n(x) = a_1(x)p_n(x+1) + a_0(x)p_n(x) + a_{-1}(x)p_n(x-1),$$

which has degree at most n , satisfies

$$\sum_{x=0}^N q_n(x)x^k \mu(x) = 0, \quad k = 0, \dots, n-1.$$

As with the classical families, *everything* related to the classical discrete polynomials can be computed explicitly, including the polynomials themselves, three-term recurrence relations, etc. For instance, the three-term recurrence relation for the Charlier polynomials $(c_n^a)_n$, normalized with leading coefficient $1/n!$, is given by

$$xc_n^a(x) = (n+1)c_{n+1}^a(x) + (n+a)c_n^a(x) + ac_{n-1}^a(x). \quad (1.15)$$

1.2 The Askey tableau. Krall and exceptional polynomials. Darboux Transforms

1.2.1 The Askey tableau

The classical and classical discrete families can be represented using (generalized) hypergeometric functions. This representation gives rise to the Askey scheme or Askey tableau: an organized hierarchy of these families of orthogonal polynomials (for a historical account see the Foreword by T. Koornwinder in [53]).